

THE HERDSMAN

Boötes



SIZE RANKING 13

BRIGHTEST STAR

Arcturus (α) -0.1

GENITIVE Boötis

ABBREVIATION Boo

HIGHEST IN SKY AT 10 PM
May–JuneFULLY VISIBLE
90°N–35°S

The Greek constellation Boötes contains the brightest star north of the celestial equator, Arcturus—Alpha (α) Boötis—which is also the fourth-brightest star in the entire sky. This large and conspicuous constellation extends from Draco and the handle of the Big Dipper (in Ursa Major) to Virgo. Faint stars in the northern part of Boötes once formed the now-defunct constellation of Quadrans Muralis, which gave its name to the Quadrantid meteor shower that radiates from this area every January.

SPECIFIC FEATURES

Arcturus is classified as a red giant, but as with most supposedly “red” stars, it actually looks orange to the unaided eye. Its coloring becomes stronger when viewed through

binoculars. In billions of years, our Sun will swell into a red giant similar to this star.

Boötes is noted for its double stars, the most celebrated of which is Izar (see p.277), or Epsilon (ϵ) Boötis, at the heart of the constellation. To the naked eye, it appears of magnitude 2.4, but high magnification on a telescope of at least 3 in (75 mm) aperture reveals a close, 5th-magnitude companion that is blue-green in color, providing one of the most beautiful contrasts of all double stars.

Much easier to divide with any small telescope are Kappa (κ) and Xi (ξ) Boötis. Kappa's stars, with components of 5th and 7th magnitudes, are unrelated, but Xi, with stars also of 5th and 7th magnitudes, is a true binary with an orbital period of 150 years and has warm yellow-orange hues.

Easiest of all are the doubles Mu (μ) Boötis, with components of 4th and 6th magnitudes, and Nu (ν) Boötis, with two 5th-magnitude components—both are widely spaced enough to divide with binoculars.

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MYTHS AND STORIES

THE BEAR-KEEPER



Boötes represents a man herding a bear (Ursa Major). Myths differ as to whether he is a hunter or a herdsman, as the constellation's brightest star, Arcturus, means “bear guard” or “bear-keeper” in Greek. The man's two dogs are represented by adjoining Canes Venatici. In Greek myth, Boötes was identified with Arcas, son of Zeus and Callisto.

ADJACENT STARS

Boötes is depicted here, leading the two hunting dogs, in an 18th-century star chart by Sir James Thornhill.

DOUBLE STAR IZAR ϵ

Epsilon (ϵ) Boötis, which is also known as Izar or Pulcherrima, is a challenging double star consisting of a bright orange star with a fainter blue-green companion star.

KITE-SHAPED CONSTELLATION ✈

Boötes, containing the bright star Arcturus, stands aloft in spring skies in the Northern Hemisphere. The crown of Corona Borealis can be seen to its left.